TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.0 ± 2.00
Correlation Stability	91.4 ± 0.92
Distribution Consistency	87.5 ± 1.96
Entropy Preservation	90.0 ± 1.10
Mean Z Score Drift	0.2 ± 0.02
Categorical Fidelity	85.4 ± 0.49
Privacy Risk	93.0 ± 0.00
Data Density	99.8 ± 0.30
Coverage Score	2.6 ± 0.66
Feature Accuracy	80.5 ± 1.80
Novelty Score	0.1 ± 0.00

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	40%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	41.298	-3.448	7.7%	OK
TB	3.299	1.490	-1.809	54.8%	Drift
DB	1.486	0.594	-0.892	60.0%	Drift
Alkphos	290.576	220.397	-70.179	24.2%	Warn
Sgpt	80.714	41.627	-39.087	48.4%	Drift
Sgot	109.911	43.416	-66.494	60.5%	Drift
TP	6.483	6.877	+0.393	6.1%	OK
ALB	3.142	3.456	+0.315	10.0%	Warn
A/G Ratio	0.947	1.003	+0.056	5.9%	OK
Selector	1.286	1.408	+0.121	9.4%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.0 ± 2.00
Correlation Stability (*2)	91.4 ± 0.92
Distribution Consistency (*3)	87.5 ± 1.96
Entropy Preservation (*4)	90.0 ± 1.10
Mean Z Score Drift (*5)	0.2 ± 0.02
Categorical Fidelity (*6)	85.4 ± 0.49
Privacy Risk (*7)	93.0 ± 0.00
Data Density (*8)	99.8 ± 0.30
Coverage Score (*9)	2.6 ± 0.66

Feature Accuracy (*10)	80.5 ± 1.80
Novelty Score (*11)	0.1 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.5529914529914537	0.6957264957264957	0.013675213675213682	
F1	0.7883981409194167	0.6377873409602444	0.7967707939089228	0.008372652989506046	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7051282051282051	0.5692307692307692	0.6735042735042736	-0.031623931623931623	
F1	0.8102825152500106	0.6741389402908468	0.7829054152565509	-0.027377099993459796	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5666666666666667	0.6923076923076923	-0.010256410256410253	
F1	0.7997903571549948	0.6587767132452338	0.7917247140751723	-0.008065643079822366	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743599	0.5948717948717949	0.5726495726495726	0.013675213675213666	
F1	0.5587432119380917	0.688307863200617	0.581363808598409	0.022620596660317314	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6170940170940172	0.6871794871794872	0.012820512820512808	
F1	0.7821343746683586	0.7231174119212261	0.7985600231850777	0.016425648516719066	

Model: DT

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6572649572649574	0.5196581196581196	0.6564102564102564	-0.00085470085470085	
F1	0.7611285138994306	0.6127409813275669	0.7528999708391169	-0.00822854306031354	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5965811965811967	0.7034188034188034	-1.1102230246251566e-17	
F1	0.8108734553482932	0.706057356605319	0.8082770894754592	-0.00259636587283409	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6307692307692309	0.6752136752136753	-0.011111111111111105	
F1	0.7971299319768493	0.7461338255365958	0.7892133108728097	-0.007916621104039711	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6914529914529914	0.7068376068376068	0	
F1	0.8279091624529537	0.8105309916326562	0.8279091624529537	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5803418803418803	0.7128205128205128	0.010256410256410275	
F1	0.8085346783591728	0.6806281974653555	0.8150781950356342	0.006543516676461292	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.9	0.2	165	26	29	8.5	4.4	1.0
2	43	Male	1.3	0.6	155	15	20	8.0	4.0	1.0
3	42	Male	16.4	8.9	245	56	87	5.4	2.0	0.5
4	18	Male	0.9	0.3	300	30	48	8.0	4.0	1.0
5	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
6	36	Female	1.2	0.4	358	160	90	8.3	4.4	1.1
7	29	Female	1.0	0.4	200	30	25	6.5	3.2	0.8
8	45	Male	2.1	0.9	180	40	35	7.0	3.5	1.0
9	38	Female	0.8	0.3	150	25	20	6.0	2.8	0.7
10	54	Male	1.5	0.7	220	60	45	7.5	3.8	1.2
11	33	Female	0.9	0.2	175	35	30	6.8	3.0	0.9
12	41	Male	3.0	1.5	250	50	40	8.0	4.0	1.1
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
2	16	Male	0.7	0.2	418	28	35	7.2	4.1	1.3
3	35	Male	1.8	0.6	275	48	178	6.5	3.2	0.9
4	38	Male	0.8	0.2	247	55	92	7.4	4.3	1.38
5	54	Male	2.2	1.2	195	55	95	6.0	3.7	1.6
6	55	Male	18.4	8.8	206	64	178	6.2	1.8	0.4
7	29	Female	1.2	0.5	210	35	45	6.5	3.0	0.8
8	45	Male	0.9	0.3	180	40	50	7.0	3.5	1.1
9	52	Female	2.5	1.0	300	50	60	6.8	3.2	0.9
10	34	Male	1.5	0.7	220	45	55	7.2	3.8	1.0
11	60	Female	0.8	0.2	150	30	40	5.5	2.5	1.2
12	27	Male	3.0	1.5	250	55	65	6.0	3.1	0.7
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	0.7	0.2	211	14	23	7.3	4.1	1.2
2	58	Female	2.8	1.3	670	48	79	4.7	1.6	0.5
3	32	Male	0.7	0.2	165	31	29	6.1	3.0	0.96
4	70	Male	3.1	1.6	198	40	28	5.6	2.0	0.5
5	55	Male	0.9	0.2	116	36	16	6.2	3.2	1.0
6	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
7	29	Female	1.2	0.5	210	22	30	6.5	3.2	1.0
8	34	Male	0.9	0.3	180	35	40	7.4	3.8	1.5
9	45	Female	2.1	1.0	250	28	50	6.8	3.5	0.9
10	60	Male	3.5	1.8	300	40	60	7.0	4.0	1.1
11	25	Female	0.7	0.2	150	20	25	6.0	2.8	0.8
12	39	Male	2.8	1.4	400	45	55	7.2	3.6	1.0
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
2	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
3	32	Male	3.7	1.6	612	50	88	6.2	1.9	0.4
4	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
5	37	Female	0.8	0.2	205	31	36	9.2	4.6	1.0
6	49	Female	0.6	0.1	185	17	26	6.6	2.9	0.7
7	29	Female	1.2	0.4	210	30	25	6.5	3.2	0.8
8	45	Male	2.5	1.0	300	50	60	7.0	3.5	1.1
9	38	Female	0.9	0.3	150	20	30	5.8	2.9	0.7
10	52	Male	3.8	1.5	400	70	80	8.0	4.0	1.2
11	60	Female	1.5	0.6	250	40	45	6.2	3.1	0.9
12	34	Male	2.0	0.8	180	25	35	7.5	3.8	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	21	Male	0.8	0.2	183	33	57	6.8	3.5	1.0
2	38	Female	0.8	0.2	145	19	23	6.1	3.1	1.03
3	50	Male	7.3	3.6	1580	88	64	5.6	2.3	0.6
4	75	Female	0.8	0.2	205	27	24	4.4	2.0	0.8
5	48	Male	5.0	2.6	555	284	190	6.5	3.3	1.0
6	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
7	34	Female	1.2	0.5	210	45	50	7.0	3.5	1.0
8	29	Male	0.9	0.3	180	30	25	6.5	3.0	1.2
9	55	Female	2.5	1.0	300	50	60	8.0	4.0	1.1
10	42	Male	1.0	0.4	220	40	35	7.5	3.8	1.2
11	60	Female	0.8	0.2	150	20	30	6.2	3.1	0.9
12	38	Male	1.5	0.7	250	60	70	8.5	4.2	1.3
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	26	Male	0.6	0.2	120	45	51	7.9	4.0	1.0
2	25	Male	0.7	0.2	185	196	401	6.5	3.9	1.5
3	45	Male	2.4	1.1	168	33	50	5.1	2.6	1.0
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	53	Female	0.7	0.1	182	20	33	4.8	1.9	0.6
6	60	Male	6.3	3.2	314	118	114	6.6	3.7	1.27
7	34	Female	1.2	0.4	210	45	50	6.5	3.5	1.1
8	29	Male	0.8	0.2	150	30	40	7.0	4.0	1.2
9	52	Female	2.5	1.0	300	60	70	7.8	3.9	0.8
10	41	Male	1.5	0.5	180	50	55	6.2	3.2	1.0
11	38	Female	0.9	0.3	220	25	30	6.0	2.8	0.9
12	45	Male	3.0	1.5	400	80	90	8.0	4.5	1.3
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	0.7	0.1	168	35	33	7.0	3.7	1.1
2	37	Male	0.8	0.2	147	27	46	5.0	2.5	1.0
3	50	Male	0.7	0.2	188	12	14	7.0	3.4	0.9
4	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
5	47	Male	0.9	0.2	192	38	24	7.3	4.3	1.4
6	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8
7	34	Male	1.2	0.4	175	45	60	7.1	3.9	1.2
8	29	Female	0.8	0.2	200	30	25	6.5	3.2	0.9
9	55	Male	1.5	0.5	250	50	70	7.8	4.0	1.1
10	42	Female	0.9	0.3	180	20	30	6.7	3.5	1.0
11	38	Male	2.1	1.0	220	60	80	8.0	4.1	1.3
12	27	Female	1.0	0.2	160	35	40	6.8	3.6	1.0
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	4.2	2.3	450	69	50	7.0	3.0	0.7
2	42	Male	30.5	14.2	285	65	130	5.2	2.1	0.6
3	39	Male	6.6	3.0	215	190	950	4.0	1.7	0.7
4	19	Male	1.4	0.8	178	13	26	8.0	4.6	1.3
5	36	Male	1.7	0.5	205	36	34	7.1	3.9	1.2
6	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
7	29	Female	0.8	0.3	210	25	30	6.5	3.2	0.8
8	55	Male	3.2	1.5	250	45	60	7.0	3.5	0.9
9	34	Female	1.1	0.4	180	20	25	6.0	2.8	0.6
10	48	Male	2.5	1.0	300	50	70	7.5	3.0	0.7
11	40	Female	0.9	0.2	190	30	40	6.8	3.4	0.9
12	62	Male	5.0	2.0	400	60	80	7.2	3.6	0.8
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
2	45	Male	3.2	1.4	512	50	58	6.0	2.7	0.8
3	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
4	30	Female	0.7	0.2	63	31	27	5.8	3.4	1.4
5	56	Female	0.7	0.1	145	26	23	7.0	4.0	1.3
6	36	Male	5.3	2.3	145	32	92	5.1	2.6	1.0
7	34	Female	1.2	0.3	210	45	50	6.5	3.2	1.0
8	29	Male	0.8	0.2	150	30	28	5.9	2.9	1.1
9	42	Female	2.5	0.8	300	60	70	7.5	3.8	1.2
10	55	Male	0.9	0.2	180	40	45	6.0	2.5	0.9
11	38	Female	1.5	0.4	220	35	33	6.8	3.4	1.1
12	50	Male	3.0	1.0	400	80	90	7.0	4.0	1.5
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 40% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	1.7	0.7	859	89	48	6.0	3.0	1.0
2	55	Female	10.9	5.1	1350	48	57	6.4	2.3	0.5
3	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
4	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4
5	40	Male	1.9	1.0	231	16	55	4.3	1.6	0.6
6	49	Female	0.8	0.2	158	19	15	6.6	3.6	1.2
7	34	Female	1.2	0.5	210	45	30	6.5	3.2	1.02
8	29	Male	0.9	0.3	180	22	25	7.0	3.6	1.1
9	50	Female	2.5	1.0	300	50	40	6.8	3.4	1.1
10	42	Male	3.0	1.2	400	60	50	7.5	3.8	1.3
11	37	Female	1.5	0.7	220	30	20	6.2	3.0	1.0
12	28	Male	0.7	0.2	160	18	15	5.9	2.8	0.9
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3